

Derivative and Integral — Topics and Subtopics

503 subtopics — THE DERIVATIVE 280 · THE INTEGRAL 213 · THE CONNECTION 10

PART A — THE DERIVATIVE

A.1 Limits and continuity

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- A.1.2 Increment of the function
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- A.1.26 Standard limit $\tan x / x \rightarrow 1$
- A.1.27 Standard limit $(1 - \cos x)/x^2 \rightarrow 1/2$
- A.1.28 Standard limit $(1 + 1/n)^n \rightarrow e$
- A.1.29 Standard limit $(1 + x)^{1/x} \rightarrow e$
- A.1.30 Standard limit $(e^x - 1)/x \rightarrow 1$
- A.1.31 Standard limit $\ln(1 + x)/x \rightarrow 1$
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- A.1.47 Intermediate value theorem
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- A.2.4 Notation $f'(x)$
- A.2.5 Notation dy/dx
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- A.2.7 Notation d/dx
- A.2.8 Notation $\dot{y}$
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- A.2.11 First principles for x , x^2 , x^3 , x^n
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PART B — THE INTEGRAL

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B.3 Table of integrals

- B.3.1 $\int x^n dx, n \neq -1$
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- B.3.3 $\int dx/x = \ln|x| + C$
- B.3.4 The exceptional case $n = -1$
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- B.3.7 $\int e^x dx$
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- B.3.12 $\int \cos x dx$
- B.3.13 $\int dx/\cos^2 x$
- B.3.14 $\int dx/\sin^2 x$
- B.3.15 $\int \tan x dx$
- B.3.16 $\int \cot x dx$
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- B.3.19 $\int dx/(x^2 + a^2)$
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B.5 The definite integral

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- B.6.32 Area under a velocity–time graph
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- B.7.1 Need for a numerical method
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- B.7.3 Trapezium rule
- B.7.4 Strips and ordinates
- B.7.5 Strip width h
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B.8 Improper integrals

- B.8.1 Infinite limit of integration
- B.8.2 Unbounded integrand
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- B.8.5 $\int(0 \text{ to } \infty) e^{(-x)} dx$
- B.8.6 Integrals through a discontinuity
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- B.9.2 Order and degree
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- B.9.6 Family of solution curves
- B.9.7 Verifying a solution
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- B.9.12 Rate proportional to a difference
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PART C — THE CONNECTION

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C.3	Fundamental theorem of calculus, part 2	C.8	Density against mass
C.4	Gradient against area	C.9	Net change as the integral of a rate
C.5	Instantaneous rate against accumulated total	C.10	Checking an integral by differentiating